

Zahra Abbasi, Ph.D.

Optics & Photonics | AI-Assisted Computational Imaging | Optical Metrology | Data-Driven Sensing

zahra.abbasizhr@gmail.com | +98 912 742 1354 | [Google Scholar](#) | [LinkedIn](#)

Profile

Ph.D. physicist with research experience in optics, Fresnel diffraction, CCD-based imaging, optical metrology, and image-based measurement. Her work includes diffraction-pattern analysis, wavefront-related optical phenomena, optical resolution enhancement, and peer-reviewed publication in *Optics Letters*. She combines a strong academic record with current QA/QC leadership experience in data-driven quality control, calibration, SPC, and laboratory measurement practices.

Research Interests

- Optics, photonics, optical metrology, interferometry, and laser-based sensing
- Fresnel diffraction, wavefront analysis, diffraction-pattern analysis, and optical measurement
- Computational imaging, computer vision, and AI-assisted analysis of optical data
- Data-driven QA/QC, calibration, SPC, measurement accuracy, and laboratory quality systems

Education

Ph.D. in Physics, Atomic and Molecular Physics 2021

Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan, Iran

Dissertation: *Fresnel Diffraction Using Plane and Divergent Wavefronts*

Supervisors: Dr. Ehsan Ahadi Akhlaghi and Dr. Ahad Saber

Evaluation: Excellent

M.Sc. in Physics, Atomic and Molecular Physics 2009

University of Zanjan, Zanjan, Iran

Thesis: *Enhancing the Resolution of a Two-Dimensional CCD Using a Photocell Displacement Method*

Supervisor: Dr. Siamak Khademi

GPA: 18.25/20

B.Sc. in Physics, Solid State Physics 2007

University of Zanjan, Zanjan, Iran

GPA: 17.67/20

Academic Honors

- Ranked 1st in B.Sc. Physics, University of Zanjan.
- Ranked 3rd in M.Sc. Physics, University of Zanjan.
- Talented Student Recognition; admitted to graduate studies through the talented-student pathway.

Publications and Presentations

Journal Publications

1. Ehsan A. Akhlaghi, Ahad Saber, and Zahra Abbasi, "Fresnel diffraction due to phase gradient singularity," *Opt. Lett.* 43, 2840-2843 (2018).
2. Khademi, S., Doroudi, A., and Abbasi, Z. **A Sub Pixel Resolution Method.** *World Academy of Science, Engineering and Technology*, 70, 2010.

Conference Presentations

1. Abbasi, Z. and Khademi, S. **Enhancing CCD resolution using two-dimensional cell displacement.** Iranian Conference on Optics and Photonics, University of Yazd, Iran, 2010.
2. Saber, A., Ahadi Akhlaghi, E., and Abbasi, Z. **Using Fresnel diffraction from a phase step to measure cylindrical wavefront curvature.** Iranian Conference on Optics and Photonics, University of Yazd, Iran, 2016.
3. Abbasi, Z., Farzanehvasht, A., Saber, A., and Ahadi Akhlaghi, E. **Using Fresnel diffraction from**

a phase step to determine spherical wavefront curvature. National Physics Conference of Iran, 2018.

4. **Abbasi, Z.**, Ahadi Akhlaghi, E., and Saber, A. **Comparison of Fourier transform and wavelet transform methods for analyzing diffraction fringes from a phase step.** 1st Conference on Optoelectronics, Applied Optics and Microelectronics (OAM), Ardabil, Iran, 2019.
5. **Abbasi, Z.**, Saber, A., and Ahadi Akhlaghi, E. **Simultaneous measurement of light wavelength and Fresnel mirror apex angle using Fresnel diffraction.** Iranian Conference on Optics and Photonics, Shahrekord, Iran, 2018.

Research Experience

Doctoral Researcher – Optics and Fresnel Diffraction

Ph.D. period

Institute for Advanced Studies in Basic Sciences (IASBS)

Investigated Fresnel diffraction, wavefront-related optical phenomena, and diffraction-pattern analysis using theoretical modeling and experimental validation, contributing to a peer-reviewed publication in *Optics Letters*.

M.Sc. Researcher – CCD Imaging and Resolution Enhancement

M.Sc. period

University of Zanjan

Developed and evaluated a CCD-based sub-pixel resolution enhancement method involving image acquisition, pixel-level measurement, and optical imaging analysis.

Selected Research Projects

Fresnel Diffraction from Phase-Gradient Singularities

Journal research project

Institute for Advanced Studies in Basic Sciences (IASBS)

- Investigated phase-gradient singularity as a physical origin of Fresnel diffraction in optical phase objects.
- Modeled diffracted intensity distributions for Fresnel double mirror and Fresnel biprism systems using Fresnel–Kirchhoff diffraction theory.
- Compared theoretical results with experimental diffraction patterns and demonstrated applications in optical metrology and simultaneous wavelength/angle measurement.

CCD-Based Sub-Pixel Resolution Enhancement and Computational Imaging

Journal

research project

University of Zanjan

- Developed a CCD-based sub-pixel resolution approach for reconstructing higher-resolution images from multiple low-resolution frames captured under fine relative motion.
- Applied MATLAB-based image reconstruction and quantitative error analysis to evaluate the enhanced image against a high-resolution reference.
- Connected optical resolution enhancement with computational imaging, image processing, and super-resolution concepts.

Fresnel Double-Mirror Optical Metrology

Conference research project

Institute for Advanced Studies in Basic Sciences (IASBS)

- Analyzed a Fresnel diffraction-based method for simultaneous measurement of incident light wavelength and Fresnel double-mirror apex angle.
- Used Fresnel–Kirchhoff diffraction modeling, fringe-spacing analysis, and camera-based intensity measurements.
- Validated the method experimentally using a He–Ne laser setup, estimating the laser wavelength at approximately 633 nm.

AI-Assisted Diffraction and Fringe Pattern Analysis

Ongoing research direction

Independent computational imaging project

- Extending prior work on Fresnel diffraction, CCD imaging, and optical pattern analysis toward AI-assisted computational imaging.
- Developing workflows for fringe-pattern interpretation, feature extraction, denoising, and parameter estimation from optical images.

Professional Experience

Quality Control Manager – Lead-Acid Battery Manufacturing 2022–Present
Tavan Pardaz Company, Iran

- Lead QC operations across lead-acid battery production, from raw material inspection to final product release.
- Supervise battery testing, including Capacity Test, CCA, OCV, Internal Resistance, and Leakage Testing, and monitor key production stages.
- Analyze test and production data, identify deviations, and maintain QC procedures, calibration practices, and technical documentation.

Technical Skills

Optics & Photonics	Fresnel diffraction; interferometry; wavefront analysis; optical metrology; laser-based sensing; CCD imaging
Computational Imaging	Image-based optical measurement; sub-pixel resolution enhancement; optical pattern and fringe analysis
Programming & Data	MATLAB; scientific computing; Fourier and wavelet analysis; numerical data analysis; visualization
AI & Computer Vision	Introductory machine learning; AI-assisted optical data analysis; image-based measurement and inspection
QA/QC & Laboratory	Data-driven QA/QC; SPC; calibration; sampling; conformity assessment; ISO/IEC 17025:2017; measurement accuracy
Documentation	Technical reporting; scientific writing; academic literature search; Microsoft Office

Teaching Experience

Lecturer / Instructor in Physics 2010–2013
Higher education institutes and universities in Iran
Taught undergraduate physics courses, including General Physics, Physics I/II, Optics, Mechanics, Mechanics Laboratory, Electricity and Magnetism, and Applied Physics.

Professional Training

- Advanced Product Quality Planning (APQP), with focus on early-stage risk identification, defect prevention, and manufacturing quality planning
- ISO/IEC 17025:2017 requirements and internal auditing for laboratories
- General quality control and quality control manager empowerment training
- Laboratory quality control requirements and regulatory supervision systems
- Conformity assessment, sampling principles, SPC, and general calibration concepts

Languages

Persian: Native | Azerbaijani Turkish: Fluent | English: Professional working proficiency